

Design and Implementation Plan

for a SaaS-Based School Management System in Liberia

Comprehensive K-12 System Design Document

Version 4.0 | April 08, 2026

Field	Details
Document Version	4.0 (WAEC/LJHSCE Corrected)
Scope	K-12 (Kindergarten through Grade 12)
Platform	Web-Based SaaS + PWA Mobile
Target Market	Republic of Liberia
Authentication	Universal Default Password + Grade Privacy Lock

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1. Liberian Education System Overview

The Liberian education system is governed by the Ministry of Education (MoE) and follows a structured progression from early childhood to secondary education.

Level	Grades	Ages	Key Details
Early Childhood	Nursery / KG	Ages 3–5	Pre-primary foundation
Lower Primary	Grades 1–3	Ages 6–8	Basic literacy & numeracy; LNAT at Grade 3
Upper Primary	Grades 4–6	Ages 9–11	LPSCE at Grade 6 (end of primary)
Junior High	Grades 7–9	Ages 12–14	LJHSCE at Grade 9 (WAEC-administered)
Senior High	Grades 10–12	Ages 15–17	WASSCE at Grade 12 (WAEC-administered)

National Examinations (WAEC-Administered)

Important: The following national examinations are NOT created, administered, or graded by individual schools. They are fully managed by the West African Examinations Council (WAEC) through the Liberia National Office. Exam papers are set by WAEC, administered at designated exam centers, and sent to WAEC/Ministry of Education for marking.

Exam	Full Name	Grade	School's Role
LJHSCE	Liberia Junior High School Certificate Examination	9th	Register candidates only
WASSCE	West African Senior School Certificate Examination	12th	Register candidates only

The school's responsibility is limited to: (1) registering eligible 9th and 12th grade students as WAEC candidates, (2) ensuring student data is accurate for exam registration, and (3) preparing students academically. The system supports this through a dedicated WAEC Exam Registration Module (see Section 12).

Grading System: Liberian schools use a letter-grade system (A = 90–100, B = 80–89, C = 70–79, D = 60–69, F = Below 60). The system will support both percentage and letter-grade display with GPA calculation.

Academic Calendar: Typically runs from September to June, with three terms/semesters. Schools may customize start dates.

2. System Architecture & Multi-Tenancy

The platform is built as a single-codebase, multi-tenant SaaS system where every school operates within an isolated data environment while sharing the same application infrastructure.

Multi-Tenancy Model

- Each school = one tenant with isolated database schema/rows
- Shared codebase, shared infrastructure, individual branding
- Tenant identified by subdomain or custom domain
- Data isolation enforced at database level (Row-Level Security)

Domain Architecture

Type	Example	Details
Default Subdomain	schoolname.eduliberia.com	Free, auto-provisioned at registration
Custom Domain	www.myschool.edu.lr	Premium tier, DNS CNAME setup
Platform Root	www.eduliberia.com	Main marketing site & school directory

Tenant Provisioning Flow

1. Proprietor/Admin registers school on the platform
2. System auto-generates subdomain (e.g., brighthorizon.eduliberia.com)
3. Landing page auto-created from school metadata (logo, colors, motto)
4. Database tenant initialized with default settings
5. Admin receives credentials to begin school configuration

3. User Roles & Permissions Matrix

The system implements role-based access control (RBAC) with 12 distinct user roles, each with carefully scoped permissions to ensure security and operational efficiency.

Role	Title	Permissions
Super Admin	Platform Owner	Full platform control, tenant management, system config, billing
Proprietor	School Owner/Investor	Read-only financial oversight, audit logs, staff monitoring, all reports
Principal	School Leader	Full school admin, staff management, academic oversight, approvals
Vice Principal	Deputy Leader	Delegated principal duties, discipline, scheduling, staff coordination
Dean of Students	Student Affairs	Student discipline, attendance oversight, student welfare, conduct records
Admin Staff	School Secretary	Enrollment processing, fee collection, record keeping, WAEC registration
IT Admin	Technical Staff	Password resets, system config, device management, technical support
Teacher	Instructor	Grades entry, attendance, lesson plans, student performance reports
Librarian	Library Manager	Book catalog, lending records, overdue tracking, library reports
Guidance Counselor	Student Support	Student counseling records, behavioral notes, parent meetings
Student	Learner	View grades (with Privacy Lock), timetable, assignments, profile
Parent/Guardian	Family	View child's grades, attendance, fee status, communicate with school

Ethics Principle: The Proprietor role is intentionally read-only. Proprietors can monitor all financial and operational data but cannot modify records, preventing conflicts of interest and ensuring staff accountability through immutable audit trails.

4. Student Authentication & Security

The system uses a low-barrier authentication model designed for Liberia's context where many students may not have personal email addresses or consistent phone access.

Universal Default Password System

- Login Credentials: Registration Number + Password
- Default Password: One universal password set by the school (e.g., SchoolName@2025)
- Password Change: Optional — students may keep the default or set their own
- Ethical Design: Students are never forced to change; the choice respects their autonomy

Authentication Flow

1. School admin registers student → System generates unique Registration Number
2. Student receives Registration Number + default password (printed slip or SMS)
3. Student visits school web portal and logs in with Reg Number + default password
4. System prompts: 'Would you like to set a personal password?' (Skip option available)
5. Student accesses dashboard — grades are protected by optional Grade Privacy Lock

Password Reset Protocol

- Self-Reset: Not available for students (prevents unauthorized access)
- IT Office Reset: Student visits IT office in person → identity verified → password reset to default
- Physical verification ensures no student can remotely hijack another's account
- IT Admin logs every reset with timestamp and verifier name for audit trail

Parent/Staff Authentication

- Parents receive account setup link via Email or SMS (school's choice)
- Staff accounts created by Admin with email-based password setup
- All non-student users can self-reset passwords via email/SMS OTP

5. Grade Privacy Lock System

Inspired by banking app security, the Grade Privacy Lock is an optional secondary authentication layer that protects a student's academic records from unauthorized viewing — even if someone else has access to their logged-in session.

How It Works

Feature	Description
Activation	Optional — student chooses to enable/disable in settings
Lock Type	4-6 digit PIN set by the student (separate from login password)
Protected Content	Grade page, GPA, transcript, exam results, academic reports
Trigger	Any attempt to open the Grades section prompts PIN verification
Session Timeout	Grades re-lock after 5 minutes of inactivity or app switch
Visual Indicator	Lock icon on Grades tab; grades hidden until unlocked
Forgot PIN	Visit IT office for in-person PIN reset (same protocol as password)

Bank-App Analogy: Just like how banking apps let you hide your balance and require fingerprint/PIN to reveal it, the Grade Privacy Lock lets students control who can see their academic performance — even on a shared device.

6. Student Enrollment & Registration Flow

Enrollment is web-only, accessible from any device (phone or computer). The system is designed for Liberia's mixed connectivity environment with offline-capable form saving.

Enrollment Process

1. Discovery: Parent/student finds school via platform directory or school's landing page
2. Application Form: Multi-step web form: personal info, guardian details, previous school, documents upload (photo, birth certificate)
3. Form Submission: Application submitted with auto-save (survives connectivity drops). Confirmation SMS/email sent.
4. Admin Review: School admin reviews application, verifies documents, approves/requests corrections
5. Fee Payment: Acceptance letter sent with fee schedule. Payment via Mobile Money (MTN, Orange) or bank transfer.
6. Registration Complete: System generates unique Registration Number. Student receives login credentials (Reg # + default password).
7. Account Active: Student logs in, optionally sets personal password, optionally enables Grade Privacy Lock.

Registration Number Format

Format: SCH-2025-0001

- SCH = 3-letter school code
- 2025 = enrollment year
- 0001 = sequential number

Unique across the entire platform. Never reused. Serves as the student's permanent identifier.

7. ID Card Generation System (NFC-Enabled)

The built-in ID card system allows schools to design, generate, and print physical NFC-enabled student ID cards directly from the platform.

Card Features

Element	Source / Details
Student Photo	Uploaded during enrollment or updated via admin portal
School Name & Logo	Auto-populated from school registration data
Student Name	Full legal name from enrollment records
Registration Number	Unique ID + barcode/QR code
Grade/Class	Current academic level and section
Academic Year	Validity period for the card
NFC Chip Data	Encrypted student ID for attendance, library, gate access

Customizable Design

- Built-in drag-and-drop card designer with templates
- Schools choose layout, colors, background, and element positioning
- Preview before printing — PDF export for professional printing
- Bulk generation for entire classes or grade levels

NFC Use Cases

- Attendance: Tap card at classroom NFC reader for automatic check-in
- Library: Tap to check out/return books
- Gate Access: Entry/exit logging for security
- Cafeteria: Meal tracking and prepaid balance (future)

8. Landing Pages & Domain Architecture

Every school on the platform gets an auto-generated, customizable landing page that serves as their public-facing website.

Auto-Generated Landing Page

- Created instantly upon school registration
- Pre-populated with school name, logo, motto, contact info, and enrollment CTA
- Responsive design — works on phones, tablets, and desktops
- SEO-optimized with school metadata

Customization Options

Section	Customizable Elements
Branding	Logo, school colors (primary/secondary), motto, banner image
Content Sections	About Us, Programs, Admissions, Calendar, News/Events, Gallery
Contact	Address, phone, email, map embed, social media links
Enrollment CTA	'Apply Now' button linking directly to enrollment form
Theme	Choose from pre-built templates or customize colors/layout

Domain Tiers

Tier	Domain	Details
Free Tier	schoolname.eduliberia.com	Auto-provisioned, instant setup
Premium Tier	www.schoolname.edu.lr	Custom domain with SSL, DNS CNAME required

9. Mobile Strategy (PWA + Future App)

Phase 1: Progressive Web App (PWA) — Launch

The primary mobile experience is a Progressive Web App, ideal for Liberia's context:

- No app store download required — works directly in the browser
- 'Add to Home Screen' creates an app-like icon on the student's phone
- Works offline for cached content (timetable, saved grades)
- Push notifications for announcements and grade updates
- Smaller data footprint than native apps
- Instant updates — no app store review delays

Phase 2: Single Branded App — “EduLiberia” (Future Growth)

When the platform achieves significant adoption, a single branded mobile app will be published:

- One app: “EduLiberia” on Google Play Store and Apple App Store
- Users search for their school within the app after download
- Reduces confusion vs. one app per school
- Leverages existing SaaS multi-tenant architecture
- Push notifications, NFC card scanning, offline sync

Why PWA First? In Liberia, data costs are high and many users have limited storage on their phones. A PWA delivers 90% of native app functionality with zero download friction.

10. Proprietor Dashboard & Ethics Module

The Proprietor role is designed around the principle of ethical oversight — providing complete visibility without the ability to manipulate data.

Dashboard Features

- Financial Overview: Real-time revenue, expenses, fee collection rates, outstanding balances, payment trends
- Staff Activity Monitor: Login times, actions performed, records modified (audit log viewer)
- Enrollment Analytics: Student count by grade, new enrollments, withdrawals, retention rates
- Academic Performance: School-wide grade averages, pass/fail rates, subject performance
- Expense Tracking: All expenditures with receipt uploads, category breakdowns, anomaly alerts
- Comparison Reports: Term-over-term and year-over-year comparisons for all metrics

Immutable Audit Trail

- Every action in the system is logged with timestamp, user, IP address, and action details
- Audit logs are append-only — no user (including Super Admin) can delete or modify them
- Proprietor has dedicated audit log viewer with search and filter
- Automatic anomaly alerts: unusual fee deletions, grade changes after deadlines, bulk record modifications

Payment Reconciliation

- Side-by-side comparison of fees collected vs. fees expected
- Mobile Money transaction verification against system records
- Alert system for discrepancies exceeding configurable thresholds
- Monthly reconciliation reports auto-generated and sent to Proprietor

11. Core Feature Modules

Academic Management

- Class scheduling and timetable generation
- Curriculum management aligned with Liberian MoE standards
- Internal exam creation, grading, and report card generation (school-level exams only)
- GPA calculation (4.0 scale) with letter grade mapping
- Transcript generation for university applications

Note: WAEC national exams (LJHSCE & WASSCE) are not created or graded by schools. See Section 12 for the WAEC Exam Registration Module.

Attendance Management

- NFC card tap for automated attendance
- Manual attendance entry by teachers (fallback)
- Real-time attendance dashboard for admin and parents
- Automated SMS/notification to parents for absences
- Attendance reports by student, class, and school-wide

Financial Management

- Fee structure setup (tuition, registration, exam, activity fees)
- Mobile Money integration (MTN MoMo, Orange Money)
- Payment tracking with receipt generation
- Installment payment plans with automated reminders
- Financial reports: income, expenses, outstanding balances

Communication Hub

- In-app messaging (teacher-parent, admin-staff, announcements)
- SMS notifications for critical updates (absences, fees, grades)
- Notice board / news feed on school landing page
- Event calendar with RSVP functionality
- Emergency broadcast system

Library Management

- Book catalog with search and categorization
- NFC-based book checkout/return
- Overdue tracking with automated notifications
- Reading history and recommendations

- Inventory management and procurement requests

Guidance & Counseling

- Confidential student counseling records
- Behavioral incident logging by Dean of Students
- Parent meeting scheduling and notes
- Academic intervention tracking
- Referral system between teachers and counselors

12. WAEC Exam Registration Module

This is a NEW dedicated section clarifying the school's role in WAEC national examinations. Schools do not create, administer, or grade WAEC exams. The school's sole responsibility is to register eligible students as examination candidates.

Background: WAEC in West Africa

The West African Examinations Council (WAEC) was established in 1952 and serves five member countries: Ghana, Nigeria, Sierra Leone, The Gambia, and Liberia (joined 1974). WAEC administers both national and international examinations across the region.

In Liberia specifically, WAEC conducts:

- LJHSCE (Liberia Junior High School Certificate Examination) — Grade 9
- WASSCE (West African Senior School Certificate Examination) — Grade 12
- LPSCE (Liberia Primary School Certificate Examination) — Grade 6 (not in this module's scope)
- LNAT (Liberia National Assessment Test) — Grade 3 (not in this module's scope)

Exam papers are set by WAEC, administered at government-designated exam centers (which may or may not be the student's school), and all answer scripts are sent to WAEC/Ministry of Education for centralized marking. Schools have zero involvement in the marking process.

What This Module Does

Feature	Description
Candidate Eligibility Check	Auto-identify 9th graders (LJHSCE) and 12th graders (WASSCE) eligible for registration
Student Data Collection	Gather required candidate info: full name, date of birth, gender, passport photo, subjects selected
Bulk Registration	Admin Staff can register all eligible students in bulk or individually
Registration Fee Tracking	Track WAEC registration fee payment per student (separate from school fees)
Data Export	Export candidate registration data in WAEC-compatible format for submission to WAEC Liberia office
Registration Status Dashboard	Track which students are registered, pending, or have issues
Exam Center Assignment	Record the exam center assigned to each student by WAEC (for reference)
Result Recording (optional)	Manually enter WAEC results when released, for school transcript records

Registration Workflow

1. System auto-flags eligible students (9th grade for LJHSCE, 12th grade for WASSCE) at start of academic year
2. Admin Staff reviews eligibility list and confirms candidates
3. Admin Staff collects WAEC registration fees from students/parents (tracked in Financial module)
4. Admin Staff enters/verifies candidate data (name, DOB, photo, subject choices for WASSCE)
5. System generates candidate list in WAEC-compatible export format
6. Admin Staff submits registration data to WAEC Liberia National Office (external process)
7. WAEC assigns exam centers and issues candidate ID slips
8. Admin Staff records exam center assignments in the system for student/parent reference
9. After exam results are released by WAEC, school can optionally record results in student transcripts

Role Permissions for WAEC Module

Role	WAEC Module Access
Admin Staff	Full access: register candidates, collect fees, export data, record results
Principal	Approve registration list, view reports
Vice Principal	View registration status, assist with verification
Teacher	View which of their students are registered
Student	View own registration status and exam center
Parent/Guardian	View child's registration status and exam center
Proprietor	Read-only: view registration statistics and fee collection

13. Liberia-Specific Considerations

Connectivity & Infrastructure

- Offline-first design: forms auto-save locally, sync when connectivity returns
- Low-bandwidth optimization: compressed images, lazy loading, minimal data transfer
- SMS fallback for critical notifications when internet is unavailable
- PWA caching for offline access to timetables and cached grades

Payment Integration

- MTN Mobile Money and Orange Money as primary payment methods
- Bank transfer support for larger institutions
- Cash payment recording (manual entry by admin with receipt)
- All payments in Liberian Dollars (LRD) with USD display option

Regulatory Compliance

- Aligned with Ministry of Education reporting requirements
- WAEC candidate registration data export for LJHSCE (9th grade) and WASSCE (12th grade)
- Student data privacy in compliance with applicable regulations
- Report card format following MoE standards
- Clear separation: internal school exams vs. WAEC national exams

Language & Accessibility

- Primary language: English (official language of Liberia)
- Simple, clear UI language suitable for varying literacy levels
- Icon-heavy navigation for intuitive use
- High-contrast mode for outdoor/bright-light phone use

14. Technology Stack & Infrastructure

Layer	Technology	Rationale
Frontend	React.js / Next.js with TypeScript	PWA-ready, component-based, responsive
Mobile	PWA (Phase 1) / React Native (Phase 2)	Cross-platform, code reuse from web
Backend	Node.js with Express / NestJS	RESTful API + WebSocket for real-time
Database	PostgreSQL with Row-Level Security	Multi-tenant isolation, ACID compliance
Authentication	JWT + Session management	Default password system + Grade Lock PIN
File Storage	Cloud Object Storage	Student photos, documents, ID card assets (S3-compatible)
Payments	MTN MoMo API, Orange Money API	Webhook-based payment confirmation
SMS Gateway	Twilio / Africa's Talking	Notifications, OTP, account setup links
Hosting	Cloud (AWS / DigitalOcean)	Auto-scaling, CDN for static assets
NFC	Web NFC API + Hardware readers	Card encoding, attendance, library

15. Implementation Roadmap

Phase 1: Foundation (Months 1–3)

- Core authentication system (Universal Default Password + Registration Numbers)
- Multi-tenant architecture with subdomain provisioning
- Basic enrollment flow (web-only application form)
- User role setup (all 12 roles with RBAC)
- Landing page auto-generation engine
- PWA shell with offline caching

Phase 2: Academic Core (Months 4–6)

- Grade management with Grade Privacy Lock
- Attendance system (manual + NFC integration)
- Timetable and scheduling module
- Report card generation (MoE-aligned format)
- Parent portal with student linking
- SMS notification integration

Phase 3: Financial & Operations (Months 7–9)

- Fee management and Mobile Money payment integration
- Proprietor Dashboard with audit trail viewer
- Payment reconciliation and anomaly alerts
- ID card designer and NFC card generation
- Library management module
- Communication hub (messaging + announcements)

Phase 4: Advanced Features (Months 10–12)

- Guidance counseling module
- Advanced analytics and reporting
- Custom domain support (premium tier)
- WAEC Exam Registration Module (LJHSCE & WASSCE candidate registration)
- Bulk operations (enrollment, grading, ID cards)
- API for third-party integrations

Phase 5: Growth — FUTURE UPDATE

- Single branded mobile app: 'EduLiberia' (Google Play + App Store)
- Biometric attendance (fingerprint)

- AI-powered academic performance predictions
- Multi-school management for education groups
- Ministry of Education reporting dashboard
- Cafeteria and transport management modules

16. Appendix

A. Registration Number Specification

Format: [SCHOOL_CODE]-[YEAR]-[SEQUENCE]

Example: BHA-2025-0042

Rules: School code = 3 uppercase letters (unique per school), Year = 4-digit enrollment year, Sequence = zero-padded 4-digit number, auto-incremented per school per year. Never reused.

B. Grade Scale Reference

Grade	Percentage	GPA	Description
A	90 – 100	4.0	Excellent
B	80 – 89	3.0	Very Good
C	70 – 79	2.0	Good
D	60 – 69	1.0	Satisfactory
F	Below 60	0.0	Fail

C. Supported Payment Methods

Method	Priority	Integration
MTN Mobile Money	Primary	USSD + API integration
Orange Money	Primary	USSD + API integration
Bank Transfer	Secondary	Manual verification by admin
Cash	Fallback	Admin records with printed receipt

D. WAEC Exam Reference

Exam	Grade	Administered By	School's Role
LJHSCE	9th	WAEC Liberia	Register candidates, collect fees
WASSCE	12th	WAEC (West Africa-wide)	Register candidates, collect fees

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End of Document